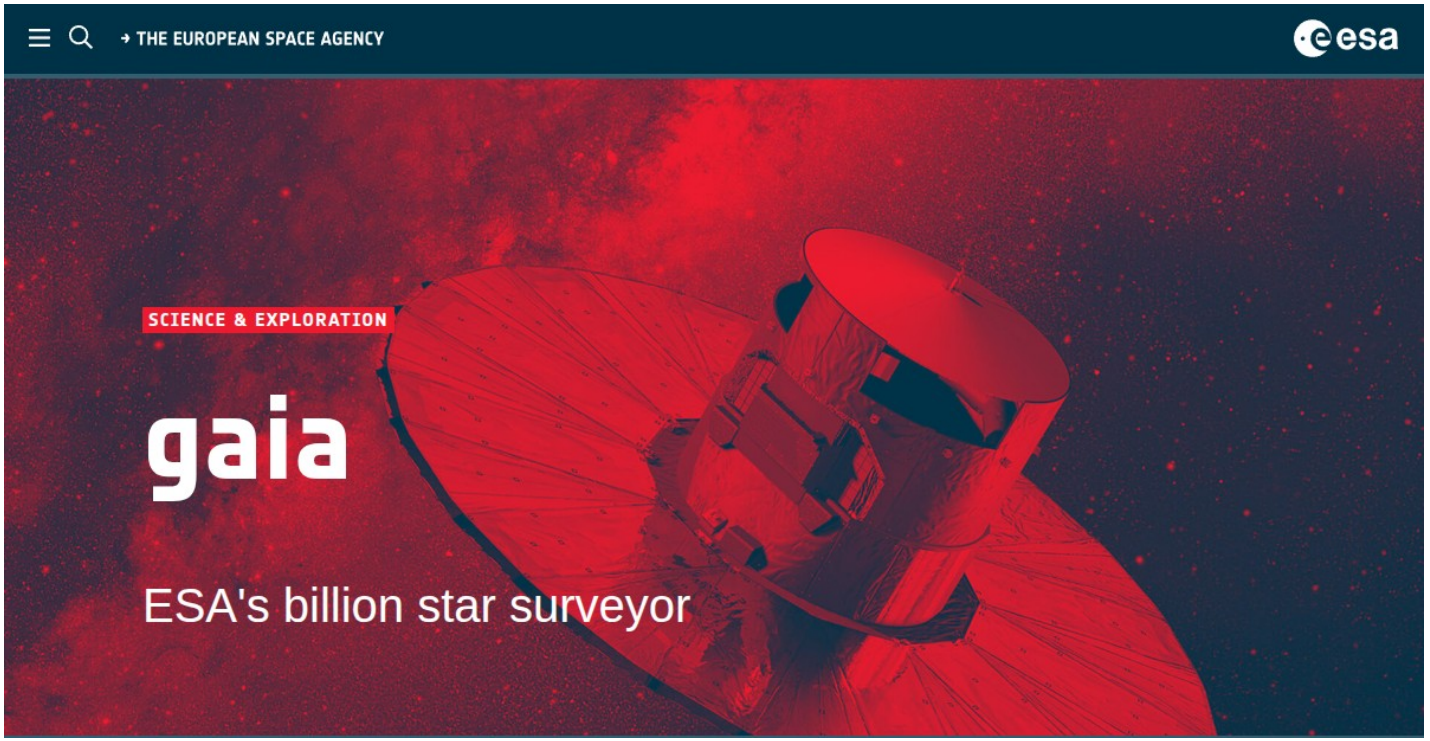
Final Project Assignment

PHYS 4730 Lab 19
Fall 2026

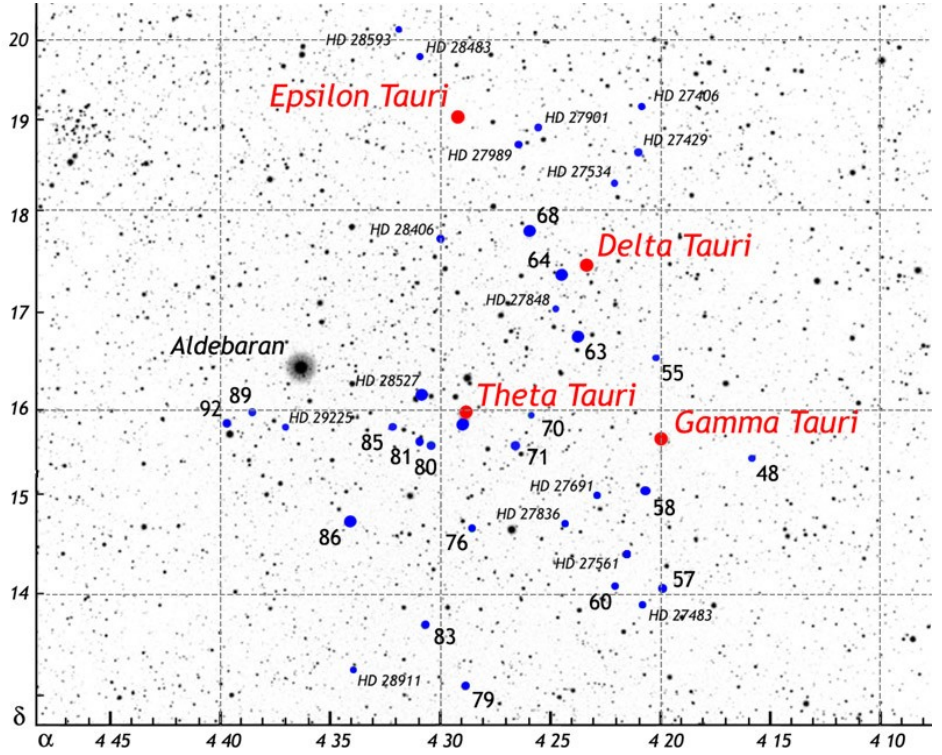
Gaia DR3 statistics project: the Hyades open cluster



- From 27 July 2014 to 15 January 2025, Gaia has made more than three trillion observations of two billion stars and other objects throughout our Milky Way galaxy and beyond, mapping their motions, luminosity, temperature and composition. Gaia's extraordinarily precise three-dimensional map will provide the data needed to tackle an enormous range of important questions related to the origin, structure and evolutionary history of our galaxy.
- Launch: 19 December 2013
- Orbit: L2 Lagrange point
- End of science observations: 15 January 2025

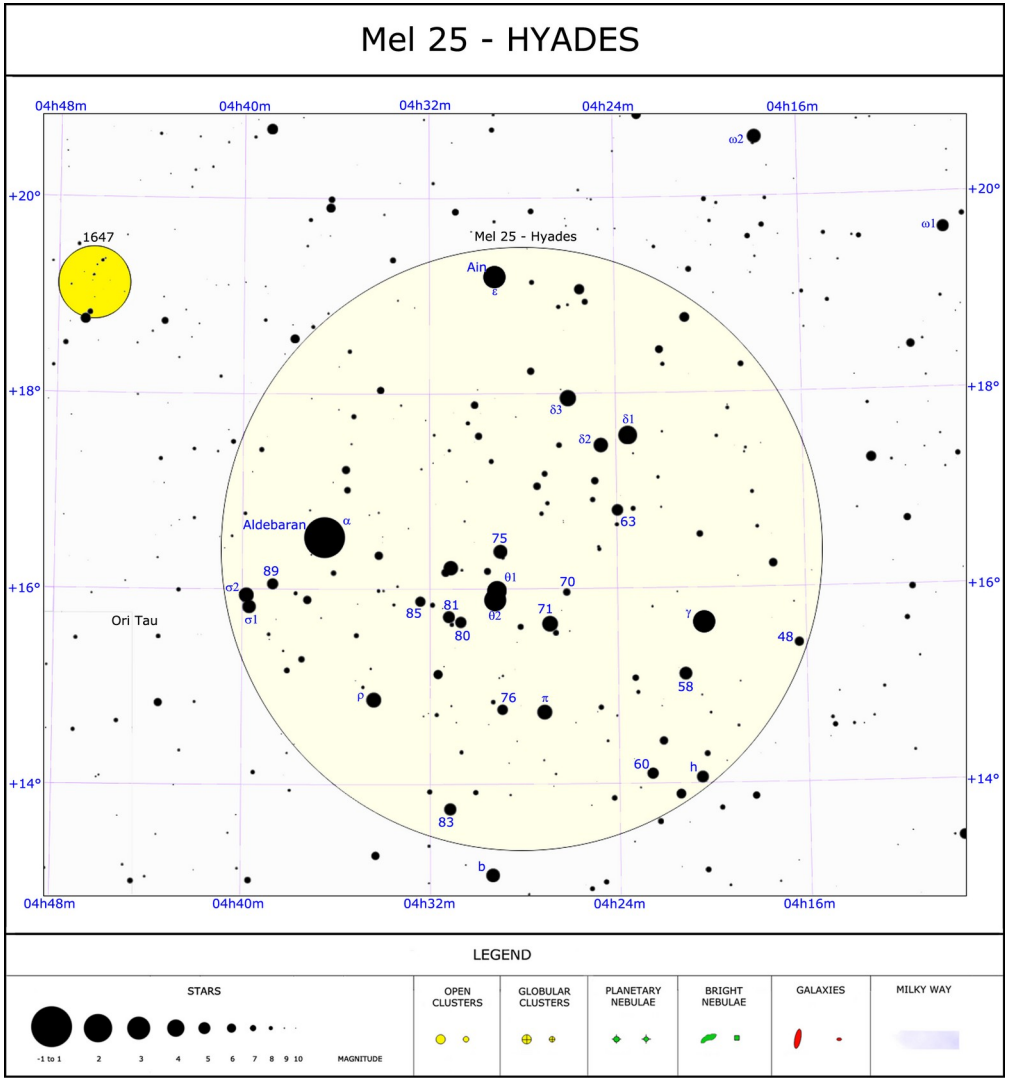
The Hyades Cluster





The Hyades Cluster

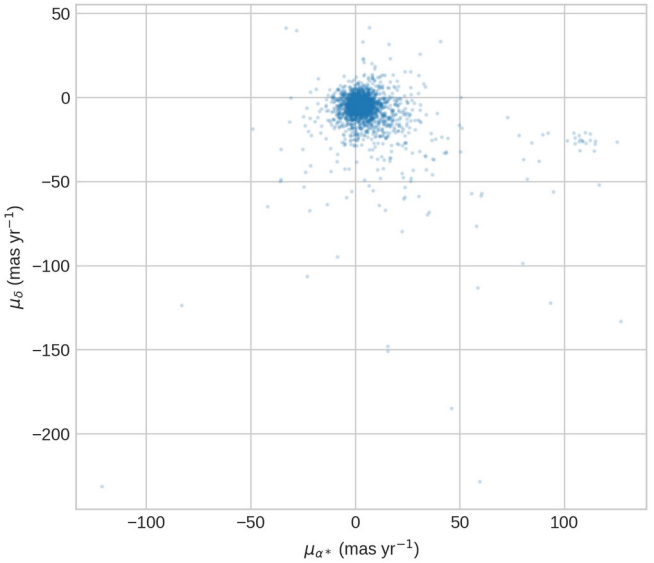


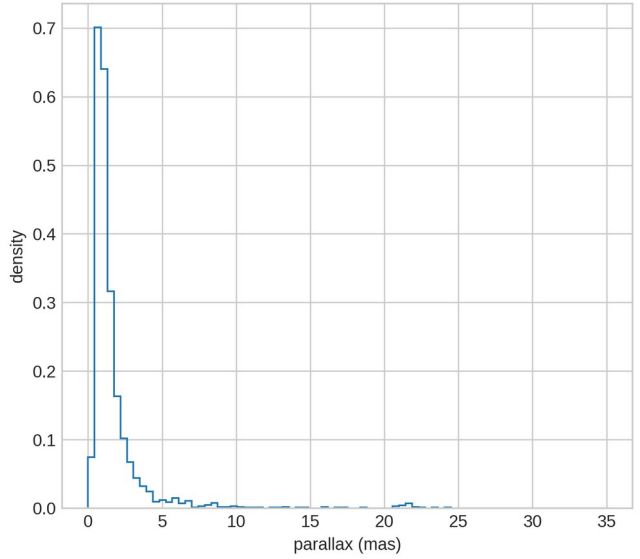


Final Project: Data

data name	description	units	symbol
`source_id`	*Gaia* identifier	—	—
`ra`	right ascension (RA)	degrees	α
`dec`	declination (DEC)	degrees	δ
`parallax`	parallax	mas[3]	ϖ
`parallax_error`	std uncertainty	mas	
`pmra`	proper motion in RA $\mu_{\alpha*} = \mu_\alpha \cos\delta$	mas/yr	$\mu_{\alpha*}$
`pmra_error`	std uncertainty	mas/yr	—
`pmdec`	proper motion in DEC	mas/yr	μ_δ
`pmdec_error`	std uncertainty	mas/yr	—
`parallax_pmra_corr`	correlation coefficient	—	—
`parallax_pmdec_corr`	correlation coefficient	—	—
`pmra_pmdec_corr`	correlation coefficient	—	—
`phot_g_mean_mag`	mean G band brightness	mag	G
`bp_rp`	*Gaia* color:	mag	$G_{\rm BP} - G_{\rm RP}$
`ruwe`	Renormalized Unit Weight Error, an astrometric fit quality indicator	—	—
`visibility_periods_used`	number of *Gaia* observing periods used in the astrometric solution	—	—

Required Analysis





- Define candidate and control samples; cut on parallax and proper motion.
- Calculate and propagate uncertainties in d, v_t.
- ML Fit of parallax data to a model with parameter s^2_{int} (intrinsic scatter).
- KS-Test candidate and control sample RVs relevant to data quality.
- Plot, analyze data in Hertzprung-Russell diagram.

Required Analysis

For each candidate star calculate it's distance from Earth in in parsecs

$$d = \frac{1}{\varpi}$$

along with its associated uncertainty.

For each candidate star calculate the *tangential speed*

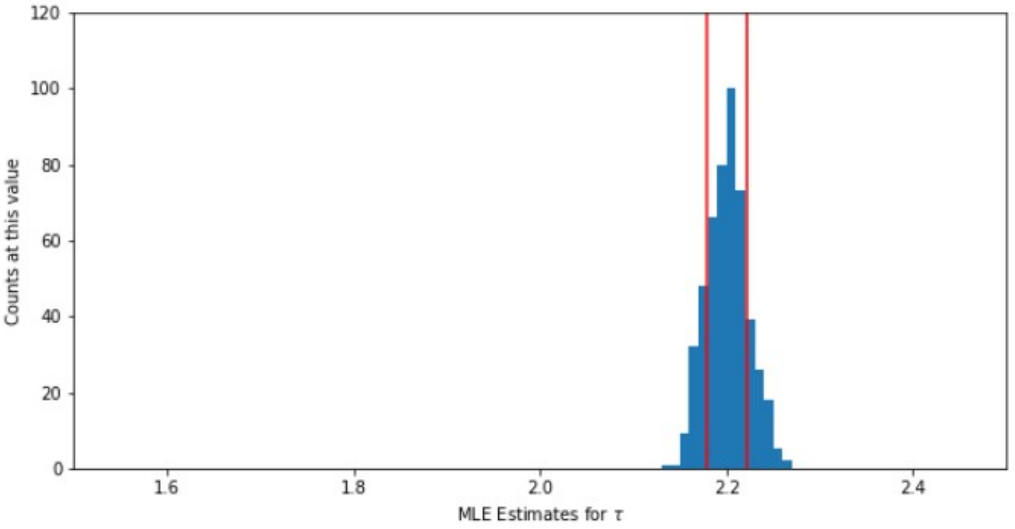
$$v_t = 4.74047 \frac{\sqrt{\mu_{\alpha*}^2 + \mu_\delta^2}}{\varpi},$$

where ϖ is in mas and proper motions are in mas/yr; v_t is in km/s. Construct the covariance matrix for $(\varpi, \mu_{\alpha*}, \mu_\delta)$ from *Gaia*'s standard errors and correlation coefficients. Use

$$\sigma_{v_t}^2 = \nabla v_t^{\mathsf{T}} C \nabla v_t.$$

- Define candidate and control samples; cut on parallax and proper motion.
- Calculate and propagate uncertainties in d, v_t.
- ML Fit of parallax data to a model with parameter s^2_{int} (intrinsic scatter).
- KS-Test candidate and control sample RVs relevant to data quality.
- Plot, analyze data in Hertzprung-Russell diagram.

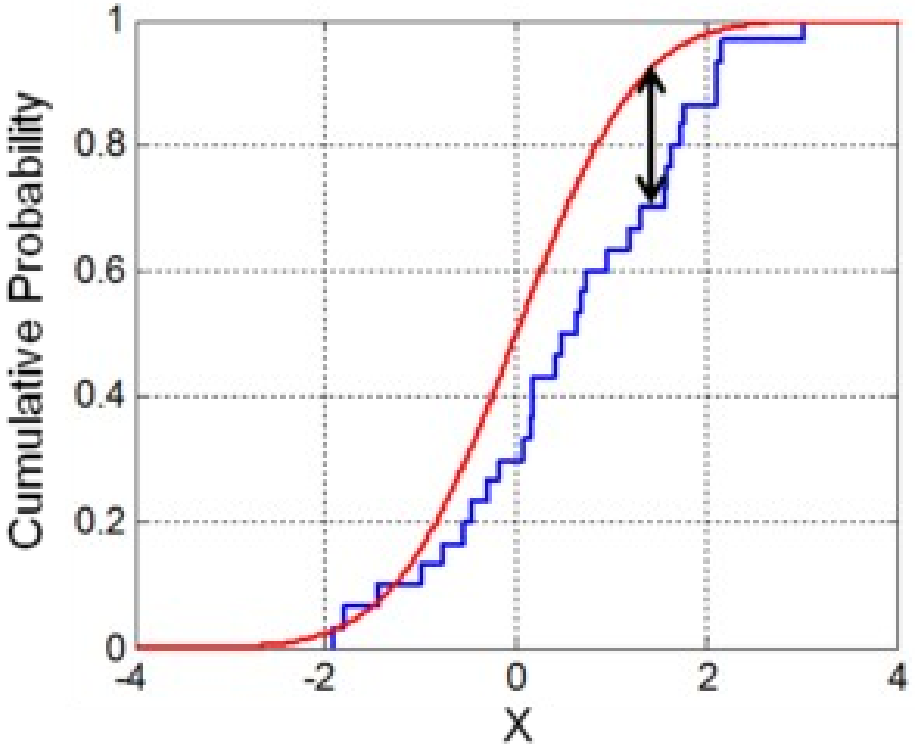
Required Analysis



Lab 10

- Define candidate and control samples; cut on parallax and proper motion.
- Calculate and propagate uncertainties in d, v_t.
- ML Fit of parallax data to a model with parameter s^2_{int} (intrinsic scatter).
- KS-Test candidate and control sample RVs relevant to data quality.
- Plot, analyze data in Hertzprung-Russell diagram.

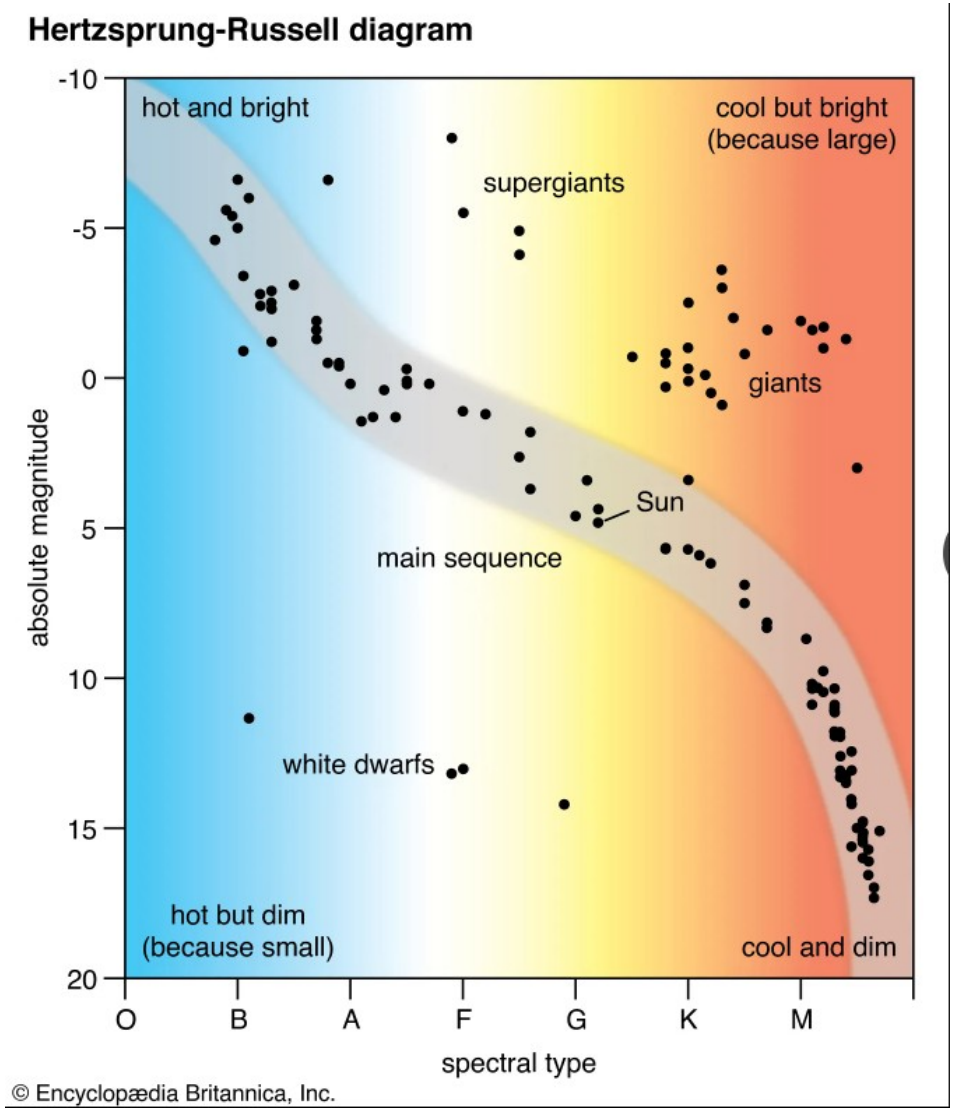
Required Analysis



Lab 16

- Define candidate and control samples; cut on parallax and proper motion.
- Calculate and propagate uncertainties in d, v_t.
- ML Fit of parallax data to a model with parameter s^2_{int} (intrinsic scatter).
- KS-Test candidate and control sample RVs relevant to data quality.
- Plot, analyze data in Hertzprung-Russell diagram.

Required Analysis



- Define candidate and control samples; cut on parallax and proper motion.
- Calculate and propagate uncertainties in d, v_t.
- ML Fit of parallax data to a model with parameter s^2_{int} (intrinsic scatter).
- KS-Test candidate and control sample RVs relevant to data quality.
- Plot, analyze data in Hertzprung-Russell diagram.

Submission

- Submit single Jupyter Notebook, with both python and markdown cells.
- By 11:59 PM on Friday Dec 11th:

  submit p4730 final-project final-project.ipynb